HEART ANATOMY - STUDY NOTES (sample document)

1. The four chambers

The heart has four chambers. The right atrium receives oxygen-poor blood from the body through the superior and inferior vena cava. The right ventricle pumps that blood to the lungs through the pulmonary valve.
The left atrium receives oxygen-rich blood from the lungs via the pulmonary veins. The left ventricle pumps oxygen-rich blood into the aorta; it has the thickest muscular wall because it must supply the entire body.

2. The heart valves

Four valves ensure one-way blood flow. The tricuspid valve lies between the right atrium and right ventricle. The mitral (bicuspid) valve lies between the left atrium and left ventricle. The pulmonary valve guards the exit of the right ventricle, and the aortic valve guards the exit of the left ventricle.
The closing of these valves produces the classic ’lub-dub’ heart sounds.

3. Path of blood through the heart

Body -> superior/inferior vena cava -> right atrium -> tricuspid valve -> right ventricle -> pulmonary valve -> pulmonary arteries -> lungs -> pulmonary veins -> left atrium -> mitral valve -> left ventricle -> aortic valve -> aorta -> body.

4. Pulmonary and systemic circulation
Pulmonary circulation carries blood between the heart and the lungs: the right ventricle pumps oxygen-poor blood to the lungs, and the pulmonary veins return oxygen-rich blood to the left atrium. Systemic circulation carries blood between the heart and the rest of the body: the left ventricle pumps oxygen-rich blood through the aorta, and the vena cavae return oxygen-poor blood to the right atrium. The two circuits operate in series, so blood passes through the heart once for every full body loop.

5. The cardiac conduction system
The heart generates its own electrical rhythm. The sinoatrial (SA) node in the right atrium is the natural pacemaker. It sends an impulse to the atrioventricular (AV) node, which delays the signal slightly so the atria can finish contracting. The impulse then travels down the Bundle of His, through the left and right bundle branches, and finally spreads through the Purkinje fibres, causing the ventricles to contract from the bottom up.

6. Basic cardiovascular physiology
Cardiac output is the volume of blood pumped per minute and equals heart rate multiplied by stroke volume. Blood pressure is roughly cardiac output times peripheral resistance. Arteries carry blood away from the heart, veins return blood to the heart, and capillaries are the thin vessels where oxygen, carbon dioxide and nutrients are exchanged with tissues.